

Agenda

- Problem Solving vs Competitive Programming
- IDE & Compiler
- Programming Language Levels
 - Machine Language
 - Assembly Language
 - High Level Language
- Program Structure
- Data Type
- Variables
- I / O
- Initialization & Declaration
- Update & Redeclaration
- Operator Types
- Comments
- Bonus Question



What Is Programming ?

In General . .

Programming refers to a technological process for telling computer which tasks to perform in order to solve problems.

★ Remember The Problem Solving Steps

- Analyze the problem**
 - Implement the algorithm**
 - Maintenance**
-



Competitive Platforms

Code Forces - LeetCode

Programmers can compete and solve in a different platforms, every platform got its purpose but these are the common one's.

The difference between each platform is that every platform is specified for something.

★ Code Forces

→ Is a platform you may attend to solve general problems and compete with others, it's problems are viewed in a form of a story that you need to explore & understand to generate the solution idea.

★ LeetCode

→ It's a platform with divided problem topics and difficulties, it's more specific to people that are preparing for an interview, it's problems are more formal & straight to the point



IDE & Compiler

★ **Definition**

→ Integrated development environment (IDE):

- ◆ It's a software application that provides comprehensive facilities to computer programmers for software development .
- ◆ it consists of(Source code – Editor – Automation tools – Debugger – Compiler)

★ **Compiler**

- It converts the written scripts (Source code) into executable program.
 - IDE's contains a Compiler In it.
-



IDE & Compiler

Types ..

There is a lot of IDE's you can use whatever you want, we will recommend IDE's and you can pick whatever you want

★ Codeblocks :

- It's an easy IDE to interact with, We recommended it.
- To download it you need to set up its environment kindly follow this [Video](#) to download it.
- Note that codeblocks is free.

★ Clion :

- It's the same as codeblocks with extended features and good looking interface.
- To download it kindly follow this link [Video](#)
- Note that Clion is not free, you can activate it with your college account for free.



Programming Language

★ Definition

→ is an agreed-upon format of symbols that allow a programmer to instruct a computer to perform certain predefined tasks.

★ Levels

→ Machine Language

→ Assembly Language

→ High - Level Language



Programming Language Levels

★ Machine Language

- Is the natural language of a computer.
- Are machine-dependent - each computer type has its own machine language.

★ Assembly Language

- It is a low-level language that allows users to write a program using alphanumeric mnemonic codes, instead of numeric code for a set of instructions examples of large assembly language programs from this time are IBM PC DOS

★ High Level Language

- Instructions are quite English-like
- Are easier to learn than machine or assembly languages.
- Have to be converted to machine languages before they can be executed using compilers

01101100
01101111
01110110
01100101



C++ Introduction

Program structure ..

First of all after creating our first (C++) Project there is a written structure that we have to understand

Example ..

- Standard Libraries
- Main Function
- Function Body
- We use #include to include a library into our code, for example the iostream library include input/output .
- Whatever we want to write or execute we have to write it in the body of the main function.
- namespace is a prefix applied to all names in the namespace set.

```
// This is a library
#include <iostream>
using namespace std;
// This is the main function
int main() {
    // this is the body of the main function
}
```




Variables

★ What is Variables . .

- In programming we define something called variables, and the variables can be defined with different keywords such as (int, float, double).
- A variable is a container that stores data values.

★ Naming Variables . .

- In (C++) the variable naming are followed by these rules:
- It can contain letters, digits & underscores
- Must begin with a letter or an underscore
- It's case sensitive which means (Var & var) are different
- It cannot contain white spaces or special characters such as (!, #, %, and more)
- Key words cannot be a variable name, it means we can't name a variable with a data type name.



Data Type

★ **Types ..**

→ Int

→ Char

→ Float

→ Double

→ Boolean

→ Long Long

→ String



Data Type

→ Int

- ◆ `int` stores whole numbers (integers) in a fixed range which is from
- ◆ -2,147,843,648 to 2,147,843,647.
- ◆ You can't store numbers with decimals (fractions) in `int`.

```
#include <iostream>
using namespace std;
int main() {
    int num = 1;
}
```



Data Type

→ char

- ◆ Stores single character between single quote.
- ◆ You can't store a full text in char.

```
#include <iostream>
using namespace std;
int main() {
    char c = 'a';
}
```



Data Type

→ Float

- ◆ Stores numbers in fractions, it can hold up to 7 decimal digits.
- ◆ It can hold a real numbers.

```
#include <iostream>
using namespace std;
int main() {
    float num = 4.5;
}
```



Data Type

→ Double

- ◆ Stores numbers in fractions, it can hold up to 15 decimal digits.
- ◆ It can hold a real numbers.

```
#include <iostream>
using namespace std;
int main() {
    double num = 2.5;
}
```



Data Type

→ Long Long

- ◆ long long stores whole numbers in a fixed range which is from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807.
- ◆ It can hold a real numbers.

```
#include <iostream>
using namespace std;
int main() {
    long long num = 4;
}
```



Data Type

→ Boolean

- ◆ boolean (bool) can take two values true or false.
- ◆ true is 1 & false is 0, it means you can bool a value with 0 or 1 / true or false.

```
#include <iostream>
using namespace std;
int main() {
    bool x = true;
    bool y = false;
    bool z = 1;
    bool d = 0;
}
```




Data Type

→ String

- ◆ string stores text, In another word a collection of characters.
- ◆ string is defined between double quote.

```
#include <iostream>
using namespace std;
int main() {
    string quote = "practice makes perfect";
}
```

Input & Output

— How to read and print . .

In (C++) to take an input (to Enter anything based on the data type) you need to use `cin>>variable`.

To print the input we took, we use `cout<<variable` by using this we can take input and provide output.

```
#include <iostream>
using namespace std;
int main() {
    int x;
    int y;
    cout<<"Enter your numbers: ";
    cin>>x>>y;
    cout<<"The summation is: "<<x+y;
}
```



```
Enter your numbers:10 5
The summation is: 15
Process finished with exit code 0
```



Initialization & Deceleration

Difference Between .

★ Deceleration :

→ Deceleration is telling the compiler that there is a container to store the data .

→ For example → `int y ;`

★ Initialization :

→ Initialization is assigning a value to the variable .

→ For example → `int y = 5 ;`



Update & Redeclaration

Difference Between .

★ Update :

→ Update is after initializing a variable you updates its value without redeclaration.

```
#include <iostream>
using namespace std;
int main() {
    int x = 5;
    x = 10;
    cout<<x;
}
```

10

Process finished with exit code 0

★ Redeclaration :

→ Redeclaration is to declare a variable with the same name twice with a different value, which gives error.

```
#include <iostream>
using namespace std;
int main() {
    int x = 5;
    int x = 10;
    cout<<x;
}
```

error: redeclaration of 'int x'

note: 'int x' previously declared here



Operators

★ **Types ..**

- Arithmetic Operators
 - Assignment Operators
 - Relational Operators
 - Logical Operators
 - Ternary Operator (Optional)
 - Bitwise Operator (Advanced)
-



Operators

★ Arithmetic Operators. .

→ Addition (+) → $10 + 10 = 20$

→ Subtraction (-) → $10 - 10 = 0$

→ Multiplication (*) → $10 * 10 = 100$

→ Division (/) → $10 / 10 = 1$

→ Modulo (%) → $10 \% 10 = 0$



Operators

★ Assignment Operators. .

→ (=) →

$X = 5$

→ (+=) →

$X += 5$ →

$X = X + 5$

→ (-=) →

$X -= 5$ →

$X = X - 5$

→ (*=) →

$X \underline{*= 5}$ →

$X = X * 5$

Operators

★ Relational Operators || Comparison Operators. .

→ Equal to (==)	→	(4 == 4)		→
(True)				
→ Not equal to (!=)	→	(5 != 4)		→
(True)				
→ Greater than (>)	→	(1 > 4)		→
(False)				
→ Less than (<)	→	(5 < 4)		→

Operators

★ Logical Operators. .

→ Logical and (&&) → (4 > 3 && 4 < 5)



→ (True)



→ Logical or (||) → (4 > 3 || 5 < 4)



❖ Logical and returns true if both operands are true.

❖ Logical or returns true if one of operands is true.

→ Logical not (!) → !(4 > 1 && 2 < 5) →

❖ Logical not returns false if statement is true.

(False)



Comments

What is the comment & what are the types?

- A comment is a text that compiler ignores, it's casually used to mark your code for a future reference
- Types are:
 - Single line comment (`//`)
 - Multi line comment (`/**/`)

Note that the output is not affected by the comments.

Example ..

```
#include <iostream>
using namespace std;
int main() {
    // this is a single line comment

    /*
    this
    is
    a
    multi line comments
    */
    int x = 5;
    cout<<x;
}
```

5

Process finished with exit code 0

Comments

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Example ..

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    // this is a single line comment

    /*
    this
    is
    a
    multi line comments
    */
    int x = 5;
    cout<<x;
}
```

5

Process finished with exit code 0

Bonus Problems

★ Problem Statement

→ Without Condition

→ Ternary



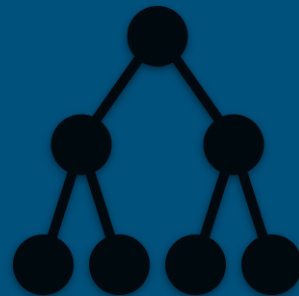


Any Questions

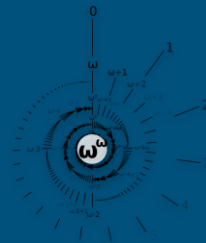




CODEFORCES



Thank You



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